

Hansika Karunathilake

Curriculum Vitae

Gampaha, Western, Sri Lanka — nostochk@gmail.com — (+94) 70 447 6354 — [LinkedIn](#) — [GitHub](#)

EDUCATION & EMPLOYMENT HISTORY

Bachelor of Science in Engineering Honours degree in Computer Engineering

Year of completion: 2026

Faculty of Engineering, University of Ruhuna

Department of Electrical and Information Engineering

OGPA: 3.8/4.0

Coursework: Computer Programming I, Computer Programming II, Introduction to Electrical Engineering, Introduction to Electronic Engineering, Mathematical Fundamentals for Engineers, Linear Algebra and Differential Equations, Analog Electronics, Data Structures and Algorithms, Electrical and Electronic Measurements, GUI Programming, Programming Project, Signals and Systems, Complex Analysis and Mathematical Transforms, Database Systems, Digital Logic Design, Problem Solving with Algorithms, Software Engineering Principles, Web Application Development, Electronic Project, Probability and Statistics, Computer Architecture and Organization, Control Systems Design, Machine Learning, Software Group Project, Discrete Mathematics, Mobile Application Development, Object Oriented Design Patterns and Principles, Computer Networks, Embedded Systems Design, Operating System and Network Programming, Numerical Methods, Artificial Intelligence, Engineering Ethics, DevOps Engineering, Advanced AI, Cloud Computing, Computer vision and image processing, Information Security, Optimization techniques for Engineers, Big Data and Analytics, Blockchain and Cyber Security, Functional Programming

Software Engineer Intern

Time Period: 2025 May - 2025 Nov

WSO2 Lanka (Pvt) Ltd

Responsibilities:

- Worked in the Integration business unit of WSO2, focusing on the development of Ballerina Connectors.
- Engineered the official Ballerina Connector for Amazon SQS, implementing core APIs and an event-driven listener for asynchronous message polling.
- Delivered 18 production APIs with 96% automated test coverage, writing 58 comprehensive unit test cases and significantly reducing manual QA effort.
- Designed and built an end-to-end AI-powered automation pipeline utilizing Anthropic LLMs to generate Ballerina connectors from OpenAPI specifications.
- Automated the sanitization of OpenAPI specifications including schema renaming, missed operationId generation, implemented AI-powered test generation, document generation, and example generation successfully presenting the full workflow to the Integration business unit and stakeholders.

PUBLICATIONS

Submitted Manuscripts

1. Karunathilake, H. et al. "Framework for blockchain-based community-assisted privacy-preserving reporting service for local governance" Submitted to the 2026 IEEE International Conference on Electrical Engineering and Emerging Technologies (ICEEET), 2026. Manuscript under review - decision expected 31 August 2026.
2. Karunathilake, H. et al. "Blockchain-based community-assisted privacy-preserving reporting service for local governance." Abstract submitted to Digital Transactions in Asia VII, 2026. Under review - decision expected 15 September 2026.

PROJECTS

Final Year Project - Framework for Blockchain-Based, Privacy-Preserving Reporting Service for Local Governance ([Link](#))

- Designed, developed and deployed the Solidity-based smart contract for public opinion polling. Orchestrated contract deployment on a private, permissioned Proof-of-Authority (PoA) Geth network.
- Developed a containerized AI oracle network utilizing pre-trained NLP and computer vision models to filter reports before on-chain storage.
- Carried out comprehensive end-to-end testing of full system including unit tests, smart contract tests, and performance testing.
- Tools/Technologies: Solidity, Go-Ethereum (Geth), Hardhat, Chai, Jest, Playwright, Python, Pytest, Docker Compose, Next.js, NestJS, IPFS, VPS/Linux

GPA Calculator for Computer Engineering Students at Faculty of Engineering, University of Ruhuna ([Link](#))

- Built a full-stack web application for engineering students to track module grades, compute semester GPA and cumulative OGPA, and check graduation and honours eligibility against curriculum requirements.
- Developed a smart elective selection optimiser that automatically finds the best combination of elective modules to maximise a student's OGPA, while satisfying credit and curriculum constraints.
- Designed a relational PostgreSQL database to model engineering curricula, module credit categories, student academic profiles, and grade records using Prisma ORM.
- Integrated secure user authentication with NextAuth.js, supporting multiple student profiles per account across different curriculum batches.
- Containerised the application with Docker for consistent and reproducible deployment.
- Tools/Technologies: Next.js, TypeScript, PostgreSQL, Prisma, NextAuth.js, Docker

CeylonTrips : Multilingual Travel & Booking Platform ([Link](#))

- Collaborated within a 3-member team to gather client requirements and architect a dynamic tour platform.
- Developed a multi-layered travel platform for a Sri Lanka-based tour company, featuring a customer-facing website, an admin dashboard, and a CMS.
- Built the backend REST API using NestJS and Prisma, handling tours (day trips, round trips, tailor-made), enquiries, media, and notifications.
- Developed the frontend using Next.js with multi-language (i18n) support, covering pages for destinations, activities, blog, and contact.
- Integrated Sanity CMS for content management and containerised the application using Docker.
- Tools/Technologies: Next.js, NestJS, TypeScript, PostgreSQL, Prisma, Sanity CMS, Docker

CERTIFICATIONS

1. Supervised Machine Learning: Regression and Classification (Coursera), DeepLearning.AI, Stanford University, Issued Aug 2024
2. Blockchain Specialization (Coursera), University at Buffalo, Issued Aug 2026

REFERENCES

- Dr. Kushan Sudheeran, Senior Lecturer, Department of Electrical and Information Engineering, University of Ruhuna, Sri Lanka. Email: kushan@eie.ruh.ac.lk
- Mr. Danesh Kuruppu, Technical Lead at WSO2, Phone: +94 77 169 0552, Email: danesh@wso2.com